

## 14.5 Partial correlation

A partial correlation measures linear association after the linear effects of conditioning variables have been removed from both variables (Definition 3.6.1, Proposition 3.6.1). Section 3.6 computed the sample version and Theorem 9.5.4 identified its square with a partial  $R^2$ ; the sampling distribution is still missing. This section shows that a partial correlation from  $n$  cases with  $q$  conditioning variables behaves exactly like an ordinary correlation from  $n - q$  cases, and that its test is the  $t$  test of a regression coefficient.

### 14.5.1 The sample partial correlation

#### Definition 14.5.1 — Sample partial correlation

Let  $\mathbf{y}, \mathbf{x} \in \mathbb{R}^n$ , let  $\mathbf{Z}$  be an  $n \times q$  matrix, and let  $\mathbf{D} = [\mathbf{1}, \mathbf{Z}]$  have rank  $q + 1 < n$ . With  $\mathbf{M}_D$  the projection onto  $\mathcal{C}(\mathbf{D})$ , put  $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{y}$  and  $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{x}$ , and suppose both are nonzero. The **sample partial correlation** of  $\mathbf{y}$  and  $\mathbf{x}$  given  $\mathbf{Z}$  is

$$r_{yx \cdot Z} = \frac{\tilde{\mathbf{y}}^\top \tilde{\mathbf{x}}}{\|\tilde{\mathbf{y}}\| \|\tilde{\mathbf{x}}\|}.$$

Both residual vectors have mean zero, because  $\mathbf{1} \in \mathcal{C}(\mathbf{D})$ , so  $r_{yx \cdot Z}$  is the ordinary sample correlation of the two residual vectors. With  $q = 0$  it is the ordinary correlation of  $\mathbf{y}$  and  $\mathbf{x}$ .

### 14.5.2 Its distribution

The key is that residualizing on  $q + 1$  columns leaves  $n - q - 1$  dimensions, exactly as centring alone leaves  $n - 1$ .

#### Theorem 14.5.1 — Distribution of a sample partial correlation

Let  $\mathbf{Z}$  be a random  $n \times q$  matrix with  $\text{rank}[\mathbf{1}, \mathbf{Z}] = q + 1$  with probability one, and  $n \geq q + 3$ . Suppose that, given  $\mathbf{Z}$ , the rows  $(y_i, x_i)$  are independent bivariate normal vectors with means  $\mathbf{a} + \mathbf{B}\mathbf{z}_{(i)}$ , affine in the  $i$ th row  $\mathbf{z}_{(i)}$  of  $\mathbf{Z}$ , and a common covariance matrix with correlation  $\rho$ . Then  $r_{yx \cdot Z}$  has the same distribution as the sample correlation of  $n - q$  independent bivariate normal pairs with correlation  $\rho$ , and it is independent of  $\mathbf{Z}$ .

In particular, if the rows  $(y_i, x_i, \mathbf{z}_{(i)}^\top)$  are independent draws from a multivariate normal distribution with a nonsingular covariance matrix, the conclusion holds with  $\rho = \rho_{yx \cdot Z}$ , the population partial correlation.

*Proof.* Fix  $\mathbf{Z}$  and write  $\mathbf{D} = [\mathbf{1}, \mathbf{Z}]$ . Let  $\mathbf{K}$  be an  $(n - q - 1) \times n$  matrix whose rows form an orthonormal basis of  $\mathcal{C}(\mathbf{D})^\perp$ , so that  $\mathbf{KD} = \mathbf{0}$  and  $\mathbf{K}^\top \mathbf{K} = \mathbf{I} - \mathbf{M}_D$ . Then

$\tilde{\mathbf{y}}^\top \tilde{\mathbf{x}} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M}_D) \mathbf{x} = (\mathbf{K}\mathbf{y})^\top (\mathbf{K}\mathbf{x})$ , and similarly for the squared lengths, so

$$r_{yx \cdot Z} = \frac{(\mathbf{K}\mathbf{y})^\top (\mathbf{K}\mathbf{x})}{\|\mathbf{K}\mathbf{y}\| \|\mathbf{K}\mathbf{x}\|}.$$

Stack the pairs as the  $n \times 2$  matrix  $[\mathbf{y}, \mathbf{x}]$ . Given  $\mathbf{Z}$ ,  $[\mathbf{y}, \mathbf{x}] = \mathbf{D}\mathbf{A} + \mathbf{E}$  for a  $(q+1) \times 2$  matrix of coefficients  $\mathbf{A}$ , where  $\mathbf{E}$  has independent  $N_2(\mathbf{0}, \mathbf{C})$  rows. Hence  $\mathbf{K}[\mathbf{y}, \mathbf{x}] = \mathbf{K}\mathbf{E}$ . Writing  $\mathbf{E} = \mathbf{E}_0 \mathbf{C}^{1/2}$  with independent standard normal entries in  $\mathbf{E}_0$ , each column of  $\mathbf{K}\mathbf{E}_0$  is  $N(\mathbf{0}, \mathbf{K}\mathbf{K}^\top) = N(\mathbf{0}, \mathbf{I}_{n-q-1})$  (Theorem 3.3.1), so the  $n-q-1$  rows of  $\mathbf{K}\mathbf{E}$  are independent  $N_2(\mathbf{0}, \mathbf{C})$  vectors. Thus, given  $\mathbf{Z}$ ,  $r_{yx \cdot Z}$  is the *uncentred* correlation of  $m = n - q - 1$  independent  $N_2(\mathbf{0}, \mathbf{C})$  pairs.

The same argument with  $q = 0$  and  $\mathbf{Z}$  absent shows that the ordinary sample correlation of  $m+1$  bivariate normal pairs is the uncentred correlation of  $m$  independent  $N_2(\mathbf{0}, \mathbf{C})$  pairs. So the conditional law of  $r_{yx \cdot Z}$  is that of an ordinary correlation from  $m+1 = n-q$  pairs, whatever  $\mathbf{Z}$  is, and Lemma 14.1.1 gives the unconditional statement.

For the multivariate normal case, Theorem 3.5.2 says that given the conditioning variables the pair is bivariate normal with mean affine in them and covariance  $\Sigma_{11 \cdot 2}$ , whose correlation is  $\rho_{yx \cdot Z}$  by Definition 3.6.1.

■

This is the result promised in [Section 3.6](#): each conditioning variable costs one case. Everything in [Section 14.4](#) transfers with  $n$  replaced by  $n-q$ : the exact law Eq. 14.4.1, the  $t$  test with  $n-q-2$  degrees of freedom, and Fisher's interval with  $\sqrt{n-q-3}$  (Fisher 1924). The conditioning variables need not be normal.

#### Example. (Twelve cases, two skewed conditioning variables)

Take  $n = 12$ ,  $q = 2$  conditioning variables drawn as squares of exponential variables, which are very skewed, a pair whose conditional means depend linearly on them, and conditional correlation  $\rho = 0.6$ . Over 100000 samples, the 10%, 50% and 90% points of the sample partial correlation are 0.257, 0.624 and 0.834. For ordinary correlations from  $n - q = 10$  normal pairs they are 0.256, 0.625 and 0.835. A two-sample Kolmogorov–Smirnov test finds no difference between the two sets of simulated values ( $p$ -value 0.09).

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

rng = np.random.default_rng(1451)
n, q, rho, reps = 12, 2, 0.6, 100_000
B = np.array([[1.0, -2.0], [0.5, 3.0]])
C = np.array([[1.0, rho], [rho, 1.0]])
pair
Lc = np.linalg.cholesky(C)

partial = np.empty(reps)
for s in range(reps):
    Zc = rng.exponential(size=(n, q)) ** 2
    pair = Zc @ B.T + rng.normal(size=(n, 2)) @ Lc.T
    D = np.column_stack([np.ones(n), Zc])
    coef, *_ = np.linalg.lstsq(D, pair, rcond=None)
```

# dependence of the pair on Z  
# conditional covariance of the pair  
# skewed conditioning variables

```

e = pair - D @ coef                                     # residuals of both variables
partial[s] = e[:, 0] @ e[:, 1] / np.sqrt((e[:, 0] @ e[:, 0]) * (e[:, 1] @ e[:, 1]))

# ordinary correlations from n - q pairs with correlation rho
x = rng.normal(size=(reps, n - q))
y = rho * x + np.sqrt(1 - rho ** 2) * rng.normal(size=(reps, n - q))
ordinary = np.array([np.corrcoef(a, b)[0, 1] for a, b in zip(x, y)])
print("quantiles, partial :", np.quantile(partial, [0.1, 0.5, 0.9]).round(3))
print("quantiles, ordinary:", np.quantile(ordinary, [0.1, 0.5, 0.9]).round(3))
print("two-sample KS p-value:", round(stats.ks_2samp(partial, ordinary).pvalue, 3))

```

[Run this code online](#)

### 14.5.3 The test of a partial correlation is a coefficient test

In the population, a zero partial correlation and a zero regression coefficient are the same thing.

#### Proposition 14.5.2 — Partial correlation and the regression coefficient

Let  $(Y, X, \mathbf{Z})$  have finite second moments, with  $\text{Cov}((\mathbf{Z}^\top, X)^\top)$  positive definite, and let  $\beta_X$  be the coefficient of  $X$  in the best linear predictor of  $Y$  from  $(\mathbf{Z}, X)$ . Then

$$\beta_X = \frac{\sigma_{yx \cdot Z}}{\sigma_{xx \cdot Z}} = \rho_{yx \cdot Z} \sqrt{\frac{\sigma_{yy \cdot Z}}{\sigma_{xx \cdot Z}}},$$

so  $\beta_X = 0$  iff  $\rho_{yx \cdot Z} = 0$ .

*Proof.* Let  $e_X = X - L(X | \mathbf{Z})$  and  $e_Y = Y - L(Y | \mathbf{Z})$  be the prediction errors of Proposition 3.6.1(a), with  $\text{Var}(e_X) = \sigma_{xx \cdot Z} > 0$  and  $\text{Cov}(e_Y, e_X) = \sigma_{yx \cdot Z}$ . The affine span of  $(\mathbf{Z}, X)$  is the orthogonal sum, in  $L^2$ , of the affine span of  $\mathbf{Z}$  and the line spanned by  $e_X$ , which is orthogonal to it. By the population version of Theorem 6.3.8, projecting  $Y$  onto the sum projects it onto each piece:  $L(Y | \mathbf{Z}, X) = L(Y | \mathbf{Z}) + (\text{Cov}(Y, e_X) / \text{Var}(e_X)) e_X$ . Since  $L(Y | \mathbf{Z})$  is orthogonal to  $e_X$ ,  $\text{Cov}(Y, e_X) = \text{Cov}(e_Y, e_X) = \sigma_{yx \cdot Z}$ . The coefficient of  $X$  in this expression comes only from the last term, because  $e_X = X -$  (affine function of  $\mathbf{Z}$ ); it is  $\sigma_{yx \cdot Z} / \sigma_{xx \cdot Z}$ . ■

This is the Frisch–Waugh–Lovell theorem in the population, and the sample version is Theorem 6.6.2 itself. Combining the two with Theorem 14.5.1:

#### Theorem 14.5.3 — The partial correlation test is the coefficient $t$ test

Regress  $\mathbf{y}$  on the  $p = q + 2$  columns  $[\mathbf{1}, \mathbf{Z}, \mathbf{x}]$ , assumed of full rank, and let  $t$  be the  $t$  statistic of the coefficient of  $\mathbf{x}$  for the value zero. Then:

- (a) for every data set,  $t = \sqrt{n-p} r_{yx \cdot Z} / \sqrt{1 - r_{yx \cdot Z}^2}$ ;
- (b) under the assumptions of [Theorem 14.5.1](#) with  $\rho = 0$ ,  $t \sim t(n-p)$ , whatever the distribution of  $\mathbf{Z}$ ;
- (c) when the rows are multivariate normal, the  $t$  test of the coefficient of  $\mathbf{x}$  and the test of  $\rho_{yx \cdot Z} = 0$  are the same test, and it has exact size.

*Proof.*

- (a) By [Theorem 9.5.4\(c\)](#) (see also Exercise (B3)),  $r_{yx \cdot Z}^2 = t^2 / (t^2 + n - p)$ . By [Theorem 6.6.2](#) the coefficient of  $\mathbf{x}$  is  $\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / \|\tilde{\mathbf{x}}\|^2$ , which has the sign of  $r_{yx \cdot Z}$ , so  $t$  and  $r_{yx \cdot Z}$  have the same sign. Solving for  $t$  gives (a).
- (b) By [Theorem 14.5.1](#),  $r_{yx \cdot Z}$  has the law of an ordinary correlation from  $n - q$  pairs with  $\rho = 0$ , and by [Proposition 14.4.1\(c\)](#) the statistic in (a) is then  $t((n - q) - 2) = t(n - p)$ .
- (c) Under multivariate normality the best linear predictor ([Theorem 2.5.2](#)) is the conditional mean ([Proposition 3.5.3](#)), so the regression coefficient of  $\mathbf{x}$  in the conditional model is  $\beta_X$  of [Proposition 14.5.2](#), which vanishes iff  $\rho_{yx \cdot Z} = 0$ . By (a) the two test statistics are monotone functions of each other, and by (b) or by [Theorem 14.1.2](#) the null distribution is exact. ■

So the test of a single coefficient in [Chapter 11](#) ([Theorem 11.6.1](#)) and the test of a partial correlation are one and the same, under either reading of the regressors. What the partial correlation adds is a scale-free measure of effect size, with its own confidence interval, which a coefficient does not provide.

#### Example. (Violent crime and urbanization)

Across the 50 states, the violent crime rate and the urbanization rate have correlation 0.345. Given the percentage of single-parent households and the poverty rate ( $q = 2$ ), their partial correlation is  $r = 0.1863$ . In the regression of violent crime on the three variables and an intercept, the coefficient of urbanization is 1.388 with  $t = 1.286$  on 46 degrees of freedom ( $p$ -value 0.2049), and indeed  $\sqrt{46} r / \sqrt{1 - r^2} = 1.286$ . With  $n - q = 48$ , the Fisher interval for  $\rho_{yx \cdot Z}$  is  $(-0.103, 0.447)$  and the exact interval is  $(-0.103, 0.443)$ . Much of the marginal association is shared with single parenthood and poverty, and what remains is compatible with none.

```
def r_cdf(c, rho, n, m=4000):
    """Exact P(r <= c) for an ordinary sample correlation of n bivariate normal
    pairs."""
    if c < 0:
        return 1 - r_cdf(-c, -rho, n, m)
    t = c * np.sqrt(n - 2) / np.sqrt(1 - c ** 2)
    if rho == 0:
        return stats.t.cdf(t, n - 2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)
    return np.mean(stats.nct.cdf(t, n - 2, rho / np.sqrt(1 - rho ** 2) * np.sqrt(V))
    )
```

```

data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
y = data["violent"].to_numpy()
x = data["urban"].to_numpy()
Z = np.column_stack([np.ones(len(y)), data["single"], data["poverty"]])
n_s, q_s = len(y), Z.shape[1] - 1 # q = 2 conditioning variables

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

ey, ex = resid(y, Z), resid(x, Z)
r_p = ey @ ex / np.sqrt((ey @ ey) * (ex @ ex)) # sample partial correlation
fit = sm.OLS(y, np.column_stack([Z, x])).fit() # p = 4 columns
t = fit.tvalues[-1]
df = n_s - 4
print(f"partial r = {r_p:.4f}, t = {t:.3f}, sqrt(df) r/sqrt(1-r^2) = "
      f"{np.sqrt(df) * r_p / np.sqrt(1 - r_p**2):.3f}")
zq = stats.norm.ppf(0.975)
lo_z, hi_z = np.tanh(np.arctanh(r_p) + np.array([-1, 1]) * zq / np.sqrt(n_s - q_s -
3))
lo_x = optimize.brentq(lambda rh: r_cdf(r_p, rh, n_s - q_s) - 0.975, -0.99, 0.99)
hi_x = optimize.brentq(lambda rh: r_cdf(r_p, rh, n_s - q_s) - 0.025, -0.99, 0.99)
print(f"Fisher z interval ({lo_z:.3f}, {hi_z:.3f}); exact ({lo_x:.3f}, {hi_x:.3f})")

```

[Run this code online](#)

*Remark* (Which variables to condition on). Conditioning can shrink an association, as in the example, reverse its sign, as in Example (Employment and unemployment) and Simpson's paradox, or create one where none exists, when the conditioning variable is a common effect of the two ([Exercise \(C1\)](#)). The distribution theory is silent on the choice of conditioning set; Chapter 25 takes it up.

## 14.5.4 Exercises

### 14.5.4.1 A. Check your understanding

#### Exercise A1.

From  $n = 40$  cases, the partial correlation of  $y$  and  $x$  given  $q = 3$  other variables is  $-0.30$ . Compute the  $t$  statistic, its degrees of freedom, and the 95% Fisher interval for the population partial correlation.

#### Exercise A2.

Explain, in terms of dimensions of subspaces, why the  $t$  test of a partial correlation given  $q$  variables has  $n - q - 2$  degrees of freedom, and why this equals  $n - p$  for the corresponding regression.

### 14.5.4.2 B. Practice

**Exercise B1.**

Let  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$ ,  $\text{rank } \mathbf{X}_1 = r_1$ ,  $\text{rank } \mathbf{X} = r$  almost surely, and suppose that, given the regressors,  $\mathbf{Y}$  follows the normal linear model with mean  $\mathbf{X}_1\boldsymbol{\beta}_1$ . Show that the partial  $R^2$  of Definition 9.5.1 has the  $\text{Beta}(\frac{r-r_1}{2}, \frac{n-r}{2})$  distribution, whatever the distribution of the regressors.

*Solution.* By Theorem 9.5.4(b),  $R_{\mathbf{X}|\mathbf{X}_1}^2 = qF/(qF+n-r)$  with  $q = r-r_1$ , where  $F$  is the  $F$  ratio for the added block. Given the regressors, the null hypothesis holds, so  $F \sim F(q, n-r)$  for every design (Theorem 14.1.2(b)), and by Lemma 14.1.1 unconditionally. Writing  $F = (U/q)/(V/(n-r))$  with independent chi-squared variables,  $R_{\mathbf{X}|\mathbf{X}_1}^2 = U/(U+V)$ , which is  $\text{Beta}(q/2, (n-r)/2)$  by Exercise (B4).

**Exercise B2.**

The *semipartial* correlation of  $\mathbf{y}$  and  $\mathbf{x}$  given  $\mathbf{Z}$  is the correlation between  $\mathbf{y}$  itself and the residual  $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{x}$ . Show that its square equals  $R_{\text{full}}^2 - R_{\text{reduced}}^2$ , the increase in  $R^2$  when  $\mathbf{x}$  is added to  $[\mathbf{1}, \mathbf{Z}]$ . How does it relate to the squared partial correlation?

*Solution.* By Lemma 9.2.1, the extra sum of squares of  $\mathbf{x}$  is  $(\tilde{\mathbf{x}}^\top \mathbf{y})^2 / \|\tilde{\mathbf{x}}\|^2$ . Dividing by  $\text{SST} = \|(\mathbf{I} - \mathbf{P}_1)\mathbf{y}\|^2$  gives  $R_{\text{full}}^2 - R_{\text{reduced}}^2$ . Since  $\tilde{\mathbf{x}}$  has mean zero,  $\tilde{\mathbf{x}}^\top \mathbf{y} = \tilde{\mathbf{x}}^\top (\mathbf{I} - \mathbf{P}_1)\mathbf{y}$ , so the quotient is the squared correlation of  $\mathbf{y}$  and  $\tilde{\mathbf{x}}$ . The squared partial correlation divides the same extra sum of squares by  $\text{SSE}_{\text{reduced}}$  instead of  $\text{SST}$ , so it equals the squared semipartial correlation divided by  $1 - R_{\text{reduced}}^2$ , and is never smaller.

**14.5.4.3 C. Going deeper****Exercise C1.**

Let  $X$  and  $Y$  be independent  $N(0, 1)$  and  $C = X + Y + U$  with  $U \sim N(0, 1)$  independent of both. Compute  $\rho_{XY.C}$ . With  $n = 200$  cases, what does the  $t$  test of the coefficient of  $X$  in the regression of  $Y$  on  $C$  and  $X$  tend to conclude? Is the conclusion about the relation between  $X$  and  $Y$  wrong, or is the question wrong?